ORCHID-SAGE

A recorded, regenerable CALPHAD assessment, shown for Cr–Ni

The record of the assessment

General protocol

the method: 45 steps, deterministic · numerical · judgement

Specific protocol

Cr–Ni: 28 inputs · 47 decisions

Run cards

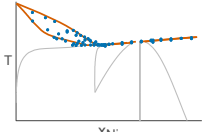
13 executions, each with its inputs, code and outputs

icons schematic

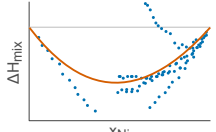
- **ORCHID** — states, σ, ids
- **PhaseForge** — MLIP one-shot
- **Liquid MD** — H_{mix}, S_{ex}
- **Physics** — checks, gaps

PhaseForge+: liquid first, then down in temperature — Cr–Ni, computed

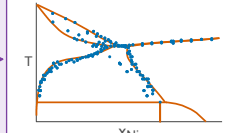
1. **Liquid**



2. **Thermochemistry**

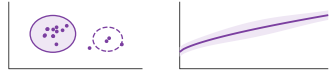


3. **Solid + exclusions**



Uncertainty — schematic

Laplace mixture of the modes; bands, invariants, topology



PI People decide

what agents may not settle alone

- rules and models, e.g. liquid first
- approvals, logged with the evidence
- second readers before records count

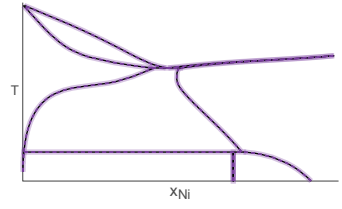
Package

schematic

TDB · dashboard · RO-Crate

Regenerated from the record

- scripted rerun — DGX Spark
- AI agent, ORCHID-SAGE MCP — Mac Studio
- original fit



Same predictions, all three

eutectic 1614 K · $CrNi_2$ top 860 K · 13/13 checks

Back to ORCHID

new records and corrections, with ids